

SIMATS Online Programming Lab

JAVA PROGRAMMING

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Q1 A college stores the names of students in an ArrayList. The administrator wants to display all student names using an Iterator instead of a for loop. 10 marks

A college stores the names of students in an ArrayList. The administrator wants to display all student names using an Iterator instead of a for loop.

Write a Java program to:

Create an ArrayList of student names.

Read 8 student names from the user.

Traverse the list using an Iterator.

Display each student name on a separate line.

Display the total number of students stored in the list.

Your Code

```
1 import java.util.ArrayList;
2 import java.util.Iterator;
3 import java.util.Scanner;
4 public class Main {
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7         ArrayList<String> names = new ArrayList<>();
8         System.out.println("Enter 8 student names:");
9         for (int i = 0; i < 8; i++) {
10             names.add(sc.nextLine());
11         }
12         Iterator<String> it = names.iterator();
13         while (it.hasNext()) {
14             System.out.println(it.next());
15         }
16         System.out.println("Total number of students: " + names.size());
17         sc.close();
18     }
19 }
```

Input

reddy
kappa
hari
dinesh
charan

Output

Enter 8 student names:
yogi
anu
jenu
reddy
kappa



Search



ENG
US



10 marks

2/3 answered

Your Code

```
1 import java.util.*;
2 public class ProductPriceAnalysis {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         ArrayList<Double> prices = new ArrayList<>();
6         System.out.println("Enter prices of 10 products:");
7         for (int i = 0; i < 10; i++) {
8             System.out.print("Price " + (i + 1) + ": ");
9             prices.add(sc.nextDouble());
10        }
11        System.out.println("\nProduct Prices:");
12        Iterator<Double> itr = prices.iterator();
13        double total = 0;
14        double highest = Double.NEGATIVE_INFINITY;
15        while (itr.hasNext()) {
16            double price = itr.next();
17            System.out.println(price);
18            total += price;
19            if (price > highest) {
20                highest = price;
21            }
22        }
23        double average = total / prices.size();
24        System.out.println("\nTotal Price = " + total);
25        System.out.println("Average Price = " + average);
26        System.out.println("Highest Price = " + highest);
27        sc.close();
28    }
29 }
```

Input

8000
9000
4000
7000

Output

Total Price = 51000.0
Average Price = 5100.0
Highest Price = 9000.0

Main.java

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         ArrayList<Integer> ids = new ArrayList<>();
6         System.out.println("Enter 10 Employee ID numbers:");
7         for (int i = 0; i < 10; i++) {
8             ids.add(sc.nextInt());
9         }
10        Iterator<Integer> it = ids.iterator();
11        while (it.hasNext()) {
12            int id = it.next();
13            if (id % 2 == 0) {
14                it.remove();
15            }
16        }
17        System.out.println("\nEmployee IDs after removing even
    IDs:");
```

Output

Enter 10 Employee ID numbers:

1
2
3
4
5
6
7
8
9
10

Employee IDs after removing even IDs:

1
3
5
7